

What are the best practices for API testing using SoapUI?

Here are some of the best API testing practices that you can follow:

1. Decide what to test

You need to adopt a risk-based testing approach for API testing. This approach helps decide what needs to be tested and when. To further simplify testing, structure your tests into individual pieces. This helps prioritise what to test and helps save time and effort.

2. Decide what NOT to test

It is equally important to know what you must NOT test. As much as possible, do not test third-party applications. Try to limit the testing to your API functionality. For instance, if you have an app that depends on a third-party payment processor, test your API to only check if it can retrieve an error message in case of some issues. These could include insufficient balance or card expiry. Once a denial message is received, it should accurately display the error message from the payment processing application.

3. Adherence to Laws and Regulations

You must ensure that the API you developed adheres to the local, national, and international laws (if required). As a professional, breaking the law can lead to fines, suspensions, and even closing down of an entire organisation. Thus, you should make it a practice to check for compliance right from the beginners' stage. Usually, laws and regulations are related to customer data privacy. Hence, ensure that storing of the data, such as customer name, credit card information, and location, etc., is done in a highly secure manner.

4. Regression Testing

As mentioned earlier, regression testing is a form of re-running API testing. It analyses the existing functional and non-functional behaviour of the API. Similarly, if there are any changes made to the application, you need to ensure that the app works as flawlessly as before. Hence, regression testing should be a compulsory and on-going process to ensure everything is working correctly.

Tips for Beginners

To simplify API testing further, here are some tips on using SoapUI for testing purposes. These tips are collected keeping in mind the common mistakes that novices usually make when using SoapUI. The testing process should become more fun, simple, and interesting after you start following these tips: